

Axler Linear Algebra

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I plan to mostly take notes on paper, but I will a lot of the exercises here because I need to practice writing $\text{\LaTeX}$ quickly and it's also a nicer format for other people to read.

1 Vector Spaces

1.A $\mathbb{R}^n$ and $\mathbb{C}^n$

These exercises are really easy but I am always scared that my proofs of these are not rigorous enough....

I do odds.

- 1 Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Proof. Let $\alpha = a + bi$ and $\beta = c + di$ where $a, b, c, d \in \mathbb{R}$.

Then we have

$$\begin{aligned}\alpha + \beta &= \\ a + bi + c + di &= \\ (a + c) + (b + d)i &= \end{aligned}$$

Because addition is commutative under the reals, we can swap the terms.

$$\begin{aligned}(c + a) + (d + b)i &= \\ c + di + a + bi &= \\ \beta + \alpha &= \end{aligned}$$

□

- 3 Show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Distribute, multiply, then rearrange terms using the commutativity of the reals to get the answer. It is quite tedious, so I won't do it here.

- 5 Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Proof. Let $\alpha = a + bi$. Then, let $\beta = -a - bi$. We can observe that

$$\begin{aligned}\alpha + \beta &= \\ a + bi - a - bi &= \\ (a - a) + i(b - b) &= 0 \end{aligned}$$

□

- 7 Show that $\frac{-1+\sqrt{3}i}{2}$ is a cube root of 1 (meaning that its cube equals 1)

Again, this is a tedious problem where you just multiply it out. Of course, I can represent this number as $e^{\frac{\pi}{3}}$ but that's cheating.

9 Find $x \in \mathbf{R}^4$ such that

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8)$$

Proof. This is vector addition, so we can apply the definition.

$$\begin{aligned}(4, -3, 1, 7) + 2x &= (5, 9, -6, 8) \\ (4 + 2x_1, -3 + 2x_2, 1 + 2x_3, 7 + 2x_4) &= (5, 9, -6, 8)\end{aligned}$$

For each term in the list, the number must hold, so in essence we get four different equations:

$$\begin{aligned}4 + 2x_1 &= 5 \\ -3 + 2x_2 &= 9 \\ 1 + 2x_3 &= -6 \\ 7 + 2x_4 &= 8\end{aligned}$$

Each of these linear equations yields only one solution, so the only solution for the equation is $x = (\frac{1}{2}, -\frac{3}{2}, -\frac{7}{2}, \frac{1}{2})$ □